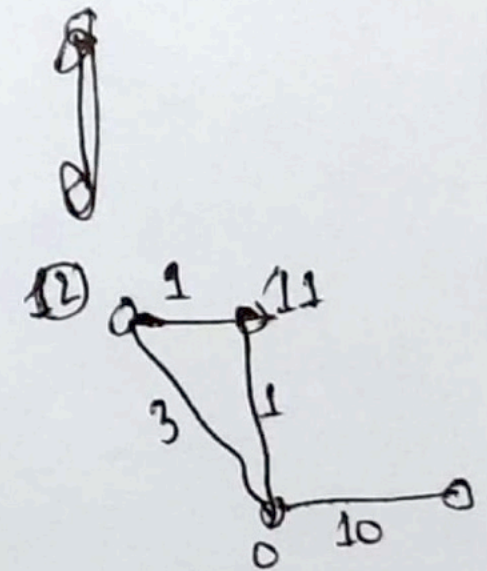
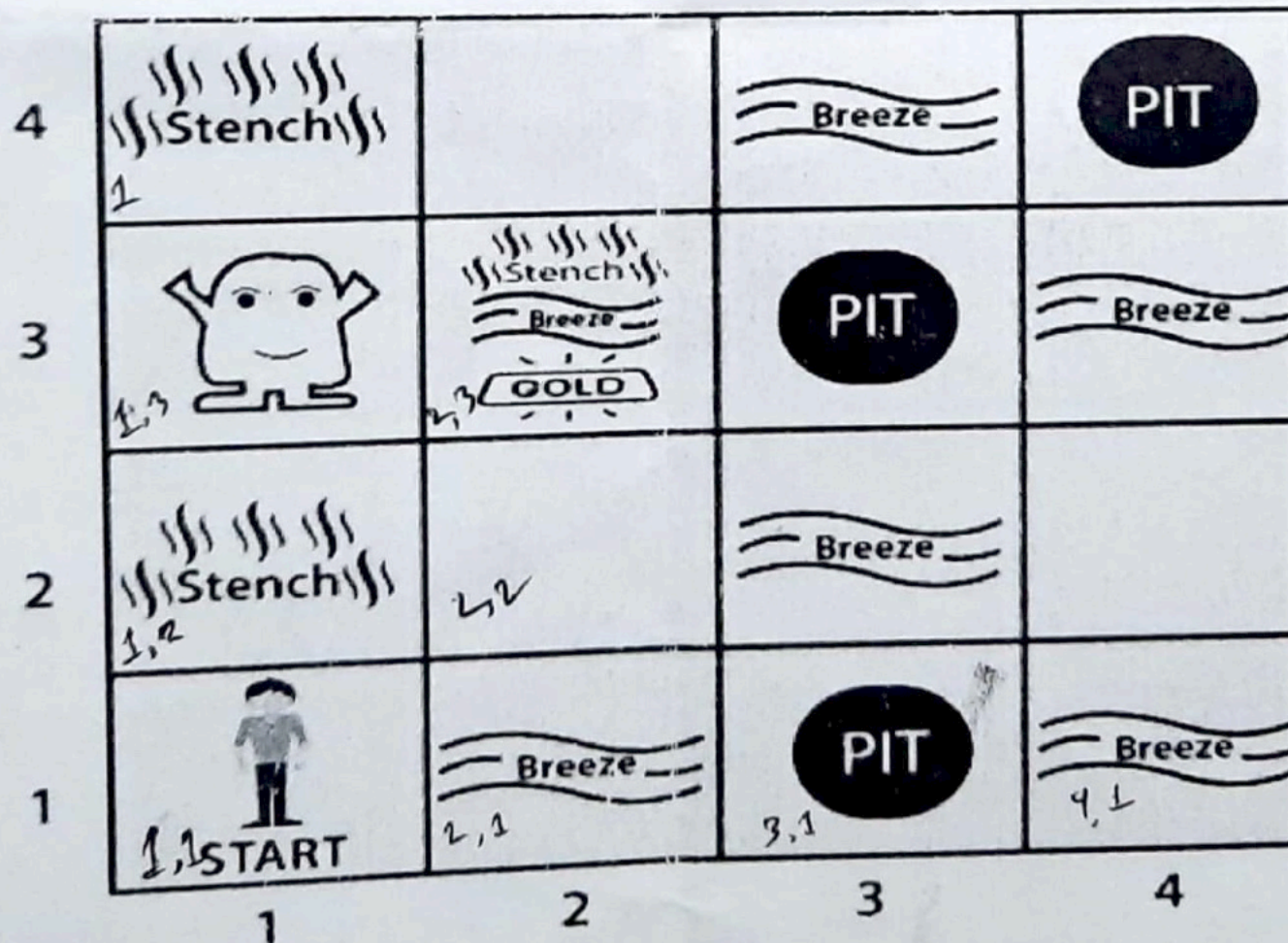


1. Prove that uniform-cost search and breadth-first search with constant step costs are optimal when used with the GRAPH-SEARCH algorithm. Compare the four evaluation criteria set of several uninformed search strategies. 5
2. What is the heuristic function of an informed search strategy? How to minimize the total estimated solution cost using the best-first search, A* search, algorithm. Show the heuristic must be admissible and consistent for the optimal solution in the A* search algorithm. 4
3. Define knowledge-based agents. Drives a propositional logic from the following Wumpus world is a cave consisting of rooms connected by passageways. Consider that the Wumpus moves from (1,3) to (2,3) as a dynamic movement. Now prove that the Wumpus is in (2,3). 6



5th Semester (Level-3, Semester-I), Midterm Examination of B.Sc. Engg.(CSE), January-June/2021, Session: 2018-19

Course Code: CTF-315 Course Title: Artificial Intelligence

Full Marks: 15 Duration: 1 hour

1. What is blind search? How to evaluate an algorithm's performance? Compare search strategies in terms of the four evaluation criteria set. 3

05

2. Write the advantage of informed search. How to make heuristic function? Define and illustrate admissible heuristic and consistent heuristic for estimating optimality of A* search algorithm. 2

05

3. Define logical agent. Write down a simple algorithm of a generic knowledge-based agent. Given a percept, the agent adds the percept to its knowledge base, asks the knowledge base for the best action, and tells the knowledge base that it has in fact taken that action. 2

03

4. What is propositional logic? Drives a propositional logic from Wumpus world is a cave consisting of rooms connected by passageways. 7

03

win - $P \vee Q$
146141 - $\neg P \vee R$

Score of work

5th Semester (Level-3, Semester-I), Midterm Examination of B.Sc. Engg.(CSE), January-June/2021, Session: 2018-19

Course Code: CIT-315 Course Title: Artificial Intelligence

Full Marks: 15 Duration: 1 hour

1. What is blind search? How to evaluate an algorithm's performance? Compare search strategies in terms of the four evaluation criteria set.
2. Write the advantage of informed search. How to make heuristic function? Define and illustrate admissible heuristic and consistent heuristic for estimating optimality of A* search algorithm.
3. Define logical agent. Write down a simple algorithm of a generic knowledge-based agent. Given a percept, the agent adds the percept to its knowledge base, asks the knowledge base for the best action, and tells the knowledge base that it has in fact taken that action.
4. What is propositional logic? Drives a propositional logic from Wumpus world is a cave consisting of rooms connected by passageways.

- (a) What are the learning outcomes of artificial intelligence?
- (b) Define in your own words the following terms: agent¹, agent² program, rationality³, autonomy⁴, deterministic, and stochastic.
- (c) What is PEAS in specifying the task environment? Illustrate and describe the structure of the model-based reflex agent.

- (a) What is uninformed search? Show that the 8-puzzle states are divided into two disjoint sets, such that no state in one set can be transformed into a state in the other set by any number of moves.
- (b) Prove that uniform-cost search and breadth-first search with constant step costs are optimal when used with the GRAPH-SEARCH algorithm.

3. (a) What is the heuristic function of informed search strategy? How to minimize the total estimated solution cost using the best-first search, A^* search, algorithm?